

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCEC504 - TRANSPORTATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The boundary till which building activities are prohibited is called _____
a) Right of way b) Boundary line c) Building line d) Control line
 - 2 The roads that connect the district headquarters to important city of other state is called _____
a) National Highway b) State Highway c) Major district road d) Other district road
 - 3 What does PPP model of highway financing stand for?
a) Public Private Property b) Partnership for Public Property
c) Partnership for Public Proceedings d) Public Private Partnership
 - 4 The minimum shoulder width recommended by IRC is _____
a) 1.0m b) 1.5m c) 2.0m d) 2.5m
 - 5 The minimum camber required in heavy rainfall area for bituminous roads as per IRC is _____
a) 1% b) 2.5% c) 2.7% d) 3%
 - 6 Trees for roadside plantation should ideally have:
A) Shallow roots and broad canopy B) Deep roots and non-shedding leaves
C) Weak stems and fast growth D) Dense foliage close to the ground
 - 7 A broken white line on the road indicates:
A) Overtaking is prohibited B) Parking is allowed
C) Lane changing is permitted when safe D) Stop line before intersection
 - 8 The STOP sign in India is shaped as:
A) Circle B) Triangle C) Rectangle D) Octagon
 - 9 The major cause of road accidents in India is:
A) Road geometry defects B) Vehicle mechanical failures
C) Human errors D) Poor visibility only
 - 10 The term "Black Spot" on a highway refers to:
A) Location with maximum vegetation B) Location of road surface failure
C) Section with frequent road accidents D) Spot where overtaking is allowed

- 11 The most effective measure to control accidents at intersections is:
 A) Increasing road width B) Providing speed breakers
 C) Installing traffic signals or roundabouts D) Road markings only
- 12 After how long can the flexible pavement be opened to the public after its construction?
 a) 24 hours b) 24 days
 c) 28 hours d) 28 days
- Q2. Solve the following.
- A) Explain in detail the factors affecting the alignment of highways 6
 B) Describe Nagpur road plan and its importance in highway development in India. 6
- Q3. Solve the following.
- A) Explain the significance of different cross-section elements in highway geometric design. Give examples of at least three elements and their functions 6
 B) Explain necessity of super-elevation and Derive its expression with suitable sketch 6
- Q4. Solve Any Two of the following.
- A) Describe the importance of aggregate shape, specific gravity, and water absorption in road construction. 6
 B) Explain penetration test and softening point test for bitumen selection 6
 C) Explain CBR test in detail and its significance in determining the suitability of soil for pavement construction 6
- Q5. Solve Any Two of the following.
- A) Explain different types of road signs used in India? Describe their shapes, colors, and significance as per IRC:67. Provide suitable examples for each category. 6
 B) Describe 3 E's that can be adopted for accident mitigation and control. 6
 C) Describe fundamental diagrams of traffic flow with neat sketch 6
- Q6. Solve Any Two of the following.
- A) Explain the steps involved in the design of rigid pavements as per IRC: 58–2011. 6
 B) Differentiate flexible pavement and rigid pavement 6
 C) Describe the procedure recommended by IRC: 37–2012 for the design of flexible pavements. Mention the role of subgrade CBR and traffic loading in the design. 6

***** End *****